

03-st-patricks-day-math-mystery

St. Patrick's Day Math Mystery: The Case of the Missing Gold — Full Content

Visual Design Notes

- **Color Palette:** Emerald green, gold/yellow, white, black accents
 - **Style:** Detective mystery meets Irish whimsy — magnifying glasses, gold coins, shamrocks, rainbow trails
 - **Layout:** Each station card is themed as a “locked door” with a keyhole graphic
 - **Fonts:** “Mystery” serif font for titles, clean sans-serif for math problems
 - **Backgrounds:** Green textured with subtle shamrock watermarks
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Storyline

The Setup (Read aloud or display on Slide 1)

🍀 BREAKING NEWS FROM SHAMROCK CITY! 🍀

Last night, the famous Shamrock City Gold Vault was robbed! The leprechaun's legendary pot of gold — worth exactly **1,847 gold coins** — has vanished!

Detective Clover has identified **5 suspects**, but she needs YOUR help to crack the case. Each suspect left behind a trail of clues hidden behind **6 locked doors**. To open each lock, you must solve the math challenges inside.

Solve all 6 locks → decode the final message → catch the thief!

The Suspects: 1. 🎩 **Finnegan O’Fiddle** — A traveling musician who was seen near the vault at midnight 2. 🌈 **Rainbow Rita** — A rainbow painter who always needs more gold paint 3. 🍄 **Mushroom Murphy** — A chef who claims he was home cooking stew 4. 📖

Emerald Emma — A librarian who says she was reading all night 5. 🐑 **Sheepish**
Shane — A shepherd whose sheep were found near the vault

Station 1: THE MULTIPLICATION LOCK

Grade 3 Version

Solve each problem. Write the letter that matches each answer on your decoder sheet.

1. $7 \times 8 = \underline{\quad}$ → If 56, letter = R
2. $9 \times 6 = \underline{\quad}$ → If 54, letter = A
3. $6 \times 7 = \underline{\quad}$ → If 42, letter = I
4. $8 \times 8 = \underline{\quad}$ → If 64, letter = N
5. $4 \times 9 = \underline{\quad}$ → If 36, letter = B

Lock Code (in order): R-A-I-N-B

Grade 4 Version

1. $23 \times 4 = \underline{\quad}$ → If 92, letter = R
2. $15 \times 6 = \underline{\quad}$ → If 90, letter = A
3. $37 \times 3 = \underline{\quad}$ → If 111, letter = I
4. $48 \times 5 = \underline{\quad}$ → If 240, letter = N
5. $64 \times 7 = \underline{\quad}$ → If 448, letter = B

Lock Code: R-A-I-N-B

Grade 5 Version

1. $234 \times 16 = \underline{\quad}$ → If 3,744, letter = R
2. $408 \times 25 = \underline{\quad}$ → If 10,200, letter = A
3. $156 \times 37 = \underline{\quad}$ → If 5,772, letter = I
4. $672 \times 14 = \underline{\quad}$ → If 9,408, letter = N
5. $395 \times 28 = \underline{\quad}$ → If 11,060, letter = B

Lock Code: R-A-I-N-B

Station 2: THE FRACTION LOCK **Grade 3 Version**

Use the fraction pictures to answer:

1. Which fraction is larger: $\frac{1}{2}$ or $\frac{1}{4}$? $\rightarrow$ Larger = O
2. What fraction of this shape is shaded? [$\frac{3}{4}$ shaded circle] $\rightarrow$ If $\frac{3}{4}$, letter = W
3. Circle the equivalent fractions: $\frac{1}{2} = \frac{?}{4}$ $\rightarrow$ If $\frac{2}{4}$, letter = R
4. What fraction is NOT shaded? [$\frac{2}{6}$ not shaded] $\rightarrow$ If $\frac{2}{6}$, letter = I
5. Put in order from least to greatest: $\frac{1}{2}$, $\frac{1}{8}$, $\frac{1}{3}$ $\rightarrow$ If $\frac{1}{8}$, $\frac{1}{3}$, $\frac{1}{2}$, letter = T

Lock Code: O-W-R-I-T**Grade 4 Version**

1. Compare: $\frac{3}{8} \bigcirc \frac{5}{8} \rightarrow$ If $<$, letter = O
2. $\frac{2}{5} + \frac{1}{5} = \underline{\hspace{1cm}}$ $\rightarrow$ If $\frac{3}{5}$, letter = W
3. Find equivalent: $\frac{3}{4} = \underline{\hspace{1cm}}/12 \rightarrow$ If $\frac{9}{12}$, letter = R
4. $\frac{7}{10} - \frac{3}{10} = \underline{\hspace{1cm}}$ $\rightarrow$ If $\frac{4}{10}$, letter = I
5. Order from least to greatest: $\frac{1}{3}$, $\frac{1}{2}$, $\frac{1}{6} \rightarrow$ If $\frac{1}{6}$, $\frac{1}{3}$, $\frac{1}{2}$, letter = T

Lock Code: O-W-R-I-T**Grade 5 Version**

1. $\frac{3}{4} + \frac{2}{3} = \underline{\hspace{1cm}}$ $\rightarrow$ If $\frac{17}{12}$ or $1\frac{5}{12}$, letter = O
2. $\frac{5}{6} - \frac{1}{4} = \underline{\hspace{1cm}}$ $\rightarrow$ If $\frac{7}{12}$, letter = W
3. $\frac{2}{3} \times \frac{3}{5} = \underline{\hspace{1cm}}$ $\rightarrow$ If $\frac{6}{15}$ or $\frac{2}{5}$, letter = R
4. Convert: $2.75 = \underline{\hspace{1cm}}$ as a fraction $\rightarrow$ If $2\frac{3}{4}$ or $\frac{11}{4}$, letter = I
5. Order: 0.5 , $\frac{3}{8}$, $\frac{7}{10} \rightarrow$ If $\frac{3}{8}$, 0.5 , $\frac{7}{10}$, letter = T

Lock Code: O-W-R-I-T

Station 3: THE DIVISION LOCK

Grade 3 Version

1. $24 \div 6 = \underline{\quad}$ → If 4, letter = A
2. $45 \div 5 = \underline{\quad}$ → If 9, letter = N
3. $36 \div 4 = \underline{\quad}$ → If 9, letter = O
4. $56 \div 8 = \underline{\quad}$ → If 7, letter = T
5. $21 \div 3 = \underline{\quad}$ → If 7, letter = E

Lock Code: A-N-O-T-E

Grade 4 Version

1. $96 \div 8 = \underline{\quad}$ → If 12, letter = A
2. $345 \div 5 = \underline{\quad}$ → If 69, letter = N
3. $252 \div 6 = \underline{\quad}$ → If 42, letter = O
4. $476 \div 7 = \underline{\quad}$ → If 68, letter = T
5. $891 \div 9 = \underline{\quad}$ → If 99, letter = E

Lock Code: A-N-O-T-E

Grade 5 Version

1. $4,284 \div 12 = \underline{\quad}$ → If 357, letter = A
2. $7,830 \div 15 = \underline{\quad}$ → If 522, letter = N
3. $9,256 \div 23 = \underline{\quad}$ → If 402 R10... (adjusted) $9,246 \div 23 = 402$, letter = O
4. $13,608 \div 36 = \underline{\quad}$ → If 378, letter = T
5. $5,472 \div 18 = \underline{\quad}$ → If 304, letter = E

Lock Code: A-N-O-T-E

Station 4: THE AREA & PERIMETER LOCK

Grade 3 Version

Find the perimeter of each shape (all measurements in shamrock units):

1. Rectangle: $6 \times 3 \rightarrow$ Perimeter = ____ $\rightarrow$ If 18, letter = F
2. Square: sides of 5 $\rightarrow$ Perimeter = ____ $\rightarrow$ If 20, letter = R
3. Rectangle: $8 \times 2 \rightarrow$ Perimeter = ____ $\rightarrow$ If 20, letter = O
4. Rectangle: $7 \times 4 \rightarrow$ Area = ____ $\rightarrow$ If 28, letter = M
5. Square: sides of 9 $\rightarrow$ Area = ____ $\rightarrow$ If 81, letter = R

Lock Code: F-R-O-M-R

Grade 4 Version

1. Rectangle: $12 \text{ cm} \times 8 \text{ cm} \rightarrow$ Area = ____ $\rightarrow$ If 96, letter = F
2. Find the missing side: Perimeter = 30, three sides are 8, 6, 9. Fourth side = ____ $\rightarrow$ If 7, letter = R
3. Composite shape (L-shape): 10×4 rectangle + 6×4 rectangle $\rightarrow$ Total area = ____ $\rightarrow$ If 64, letter = O
4. Square with perimeter of 48 ft $\rightarrow$ Side length = ____ $\rightarrow$ If 12, letter = M
5. Rectangle area = 72, width = 8 $\rightarrow$ length = ____ $\rightarrow$ If 9, letter = R

Lock Code: F-R-O-M-R

Grade 5 Version

1. Parallelogram: base 14 cm, height 8 cm $\rightarrow$ Area = ____ $\rightarrow$ If 112, letter = F
2. Triangle: base 20 in, height 9 in $\rightarrow$ Area = ____ $\rightarrow$ If 90, letter = R
3. Composite: Rectangle 15×10 minus square $5 \times 5 \rightarrow$ Area = ____ $\rightarrow$ If 125, letter = O
4. Room is $18 \text{ ft} \times 12 \text{ ft}$. Carpet costs \$3 per sq ft. Total cost = ____ $\rightarrow$ If \$648, letter = M
5. Perimeter of triangle: sides 13.5, 8.25, 10.75 $\rightarrow$ Perimeter = ____ $\rightarrow$ If 32.5, letter = R

Lock Code: F-R-O-M-R

Station 5: THE WORD PROBLEM LOCK

Grade 3 Version

1. Finnegan found 24 gold coins on Monday and 3 times as many on Tuesday. How many did he find on Tuesday? → If 72, letter = I
2. Rita had 156 gold paint jars. She used 89. How many are left? → If 67, letter = T
3. Murphy made 8 pots of stew. Each pot serves 6 people. How many people can eat? → If 48, letter = A
4. Emma has 42 books to put equally on 6 shelves. How many per shelf? → If 7, letter = H
5. Shane has 135 sheep. 47 are white, 63 are brown. The rest are black. How many are black? → If 25, letter = E

Lock Code: I-T-A-H-E (unscramble to: ??? — see decoder)

Grade 4 Version

1. The vault had 1,847 coins in bags of 25. How many FULL bags? How many left over? → If 73 R22, letter = I
2. Rita bought 3.5 gallons of paint at \$12 per gallon. Total cost? → If \$42, letter = T
3. Murphy used $\frac{2}{3}$ of his 18 potatoes. How many did he use? → If 12, letter = A
4. Emma read for 45 minutes per day for 2 weeks. Total minutes? → If 630, letter = H
5. Shane's field is 24 yards by 18 yards. What's the perimeter of fencing needed? → If 84 yards, letter = E

Lock Code: I-T-A-H-E

Grade 5 Version

1. The vault's coins weigh 0.25 oz each. Total weight of 1,847 coins? → If 461.75 oz, letter = I
2. Rita mixed paint: $2\frac{1}{3}$ cups red + $1\frac{3}{4}$ cups yellow. Total? → If $4\frac{1}{12}$ cups, letter = T
3. Murphy's stew recipe serves 8. He needs to serve 52. What fraction of the original amount of each ingredient does he need? Express as a mixed number. → If $6\frac{1}{2}$ (times the recipe), letter = A

4. Emma cataloged 2,340 books in 12 days, same number each day. On day 13 she did half. Total books after 13 days? → If 2,437.5 → adjusted: 2,340 in 12 days = 195/day. Half = 97.5 → Use: 2,436 books in 12 days = 203/day. Half = 101.5 → **Adjusted:** Emma cataloged books for 12 days. She did 195 per day for 10 days and 210 per day for 2 days. Total? → If 2,370, letter = H
5. Shane's rectangular field has an area of 1,296 sq yards. If the length is 48 yards, what's the width? → If 27, letter = E

Lock Code: I-T-A-H-E

Station 6: THE DATA LOCK

Grade 3 Version

Use the bar graph: "Gold Coins Found by Each Suspect" - Finnegan: 15, Rita: 30, Murphy: 10, Emma: 5, Shane: 20

1. Who found the most coins? → If Rita, letter = R
2. How many more did Rita find than Murphy? → If 20, letter = I
3. How many coins total? → If 80, letter = T
4. What is the difference between the most and least? → If 25, letter = A
5. If Shane found 5 more, how many would he have? → If 25, letter = !

Lock Code: R-I-T-A-!

Grade 4 Version

Use the table: "Suspect Activity Log (Hours)"

Suspect	Monday	Tuesday	Wednesday
Finnegan	3	5	4
Rita	6	7	8
Murphy	2	2	3

Suspect	Monday	Tuesday	Wednesday
Emma	4	4	4
Shane	5	3	1

1. Who spent the most TOTAL hours on activities? → If Rita (21), letter = R
2. What is the mean (average) of Murphy's hours? → If 2.33... round to ~2, letter = I
(use: What is the range of Shane's hours? If 4, letter = I)
3. How many total hours for ALL suspects on Monday? → If 20, letter = T
4. Who had the same hours every day? → If Emma, letter = A
5. How many more total hours did Rita have than Shane? → If $21-9=12$, letter = !

Lock Code: R-I-T-A-!

Grade 5 Version (More complex data interpretation)

Use the double bar graph and table provided showing “Coins Found” vs “Coins Spent” per suspect.

1-5: Similar structure leading to **R-I-T-A-!**

THE DECODER

Students combine all 6 lock codes: - Station 1: **RAINB** - Station 2: **OWRIT** - Station 3: **ANOTE** - Station 4: **FROMR** - Station 5: **ITAHE** (unscramble within station) - Station 6: **RITA!**

Combined message: “RAINBOW RITA NOTE FROM RITA HE— RITA!”

Final reveal: Rearranging the clues: “**RAINBOW RITA**” wrote a note that says “**THE GOLD IS MINE!**” → **Rainbow Rita is the thief!** 🌈

Students fill in their “Case Solved” certificate: “I solved The Case of the Missing Gold! The thief was **Rainbow Rita!**”

“Case Solved!” Certificate



This certifies that _____

has successfully solved THE CASE OF THE MISSING GOLD!

By demonstrating outstanding skills in: ☐ Multiplication ☐ Division ☐ Fractions ☐ Area & Perimeter ☐ Word Problems ☐ Data Analysis

The thief was: _____

Date: _____ **Detective Badge #:** _____

Google Slides Interactive Version Outline

Slide 1: Title — Dark green background with gold coins animation

Slide 2: “The Crime Scene” — Story setup with animated suspect lineup

Slide 3: Instructions — How to play, rules

Slides 4-9: Station 1 (one slide per problem, click to reveal if correct)

- Each problem appears on a “locked door” graphic
- Students type answer in text box
- “Check” button links to answer reveal slide ### Slides 10-15: Station 2 (same format) ### Slides 16-21: Station 3 ### Slides 22-27: Station 4 ### Slides 28-33: Station 5 ### Slides 34-39: Station 6 ### Slide 40: Decoder page — students type their letters ### Slide 41: “THE REVEAL!” — Click to unmask the thief (animated gold coins fall) ### Slide 42: Certificate — Students type their name

Interactive features: - Clickable “lock” buttons that link to next clue - Typed answer boxes - Linked navigation (can’t skip ahead) - Self-checking where possible (color-coded answer reveals)

Teacher Setup Guide

Preparation (15 min)

1. Print station cards (color recommended) and cut
2. Optional: laminate for reuse
3. Set up 6 stations around the room OR distribute as packet
4. Print student answer sheets (1 per student or pair)
5. Choose difficulty level per student/group

Classroom Options

- **Centers rotation:** 6 stations, 8-10 min each
- **Whole class:** Project one station at a time, all solve together
- **Partner work:** Print packets, pairs work through all 6
- **Digital:** Assign Google Slides version

Time: 45-60 minutes total